4DVarNet inference speed

Is the query time a problem? — measured: 0.17 ms per frame, 5,725x real time

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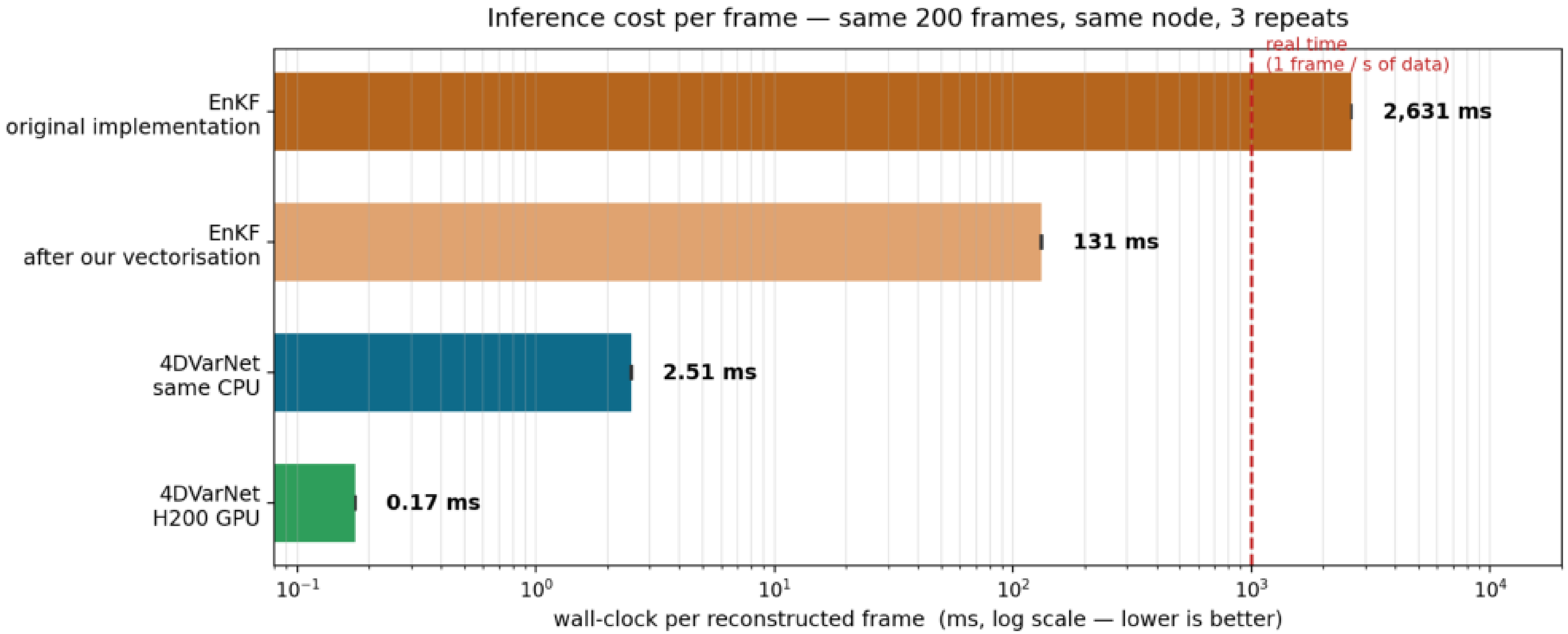
Inference cost per frame



INTEL(R) XEON(R) PLATINUM 8562Y+ · 16 cores · NVIDIA H200 · node gpu52. 200 frames of atc-20130811; methods measured one at a time, cycling through all of them each round rather than finishing one before starting the next; error bars = std over 3 rounds (max spread 2.0%). CPU rows are the like-for-like comparison; the EnKF has no GPU path (its Kalman update is host numpy), so the GPU row is a deployment figure, not a head-to-head.

- On an H200: 0.17 ms/frame = 5,725 frames per second = 5,725x real time
- On the SAME CPU: 52x faster than the EnKF (2.51 vs 131 ms) — this is not a GPU effect
- The EnKF's original implementation is slower than real time (2.6 s to process 1 s of data)

The EnKF's cost is its implementation, not its algorithm

- Profiling one assimilation step — where the time goes:
  - 96%   building the localization matrix — a table of distance weights
  - 2%   the Kalman update itself, the actual filtering

- We vectorised that one function: 20x faster (2,631 -> 131 ms/frame), output bit-identical.
- So the comparison uses 131 ms, not 2,631 ms. The original project was not modified.

Reconstruction accuracy — held-out test set

- 7 held-out days (2013-08-11 to 2013-09-29), 276,800 frames the models never trained on. Same days, same frames and the same physical bounds for every row. RMSE over the whole state — the convention the EnKF project uses for its own filter.

	4DVarNet	EnKF
every 4th frame observed	0.189	0.243
every frame observed	0.170	still running

Where this goes next

- Done: denser observations
  - moved from observing every 4th frame to every frame — RMSE 0.189 -> 0.170 on the held-out days, 19% lower error
  - costs nothing at inference — the solver's runtime is unchanged
- Next: more model capacity
  - the GENN prior currently has 13,900 parameters
  - the paper's Lorenz-96 setup uses roughly 50,000 — there is room